

through production or turns into an expensive batch of rejects. Rather than repeating familiar textbook advice, it is worth examining the less obvious issues that frequently create manufacturing problems.

PCB size constraints. A PCB design may need to be compact, but pick-and-place machines use conveyor belts that physically grip the board. This typically requires at least 5mm of clearance on two opposite edges that are free from SMD components. Designs that provide only about 2mm clearance often require manual intervention during assembly, slowing production and increasing cost.

Heavy components. When components are placed on both sides of a PCB, the second reflow pass reheats the solder joints on the first side. If heavy components are mounted on that side, the solder can re-melt and the parts may drop off the board. Wherever possible, heavier components should be placed on the first side of the board; otherwise, a custom jig fixture may be required to secure them during reflow.

Fiducial marks. These small golden dots are not decorative elements. They serve as positioning references for pick-and-place machines, allowing the system to accurately align the PCB before placing components. Without them, automated placement systems cannot reliably determine board orientation or alignment.

Fiducials, apertures in PCB assembly

Are you paying enough attention to your fiducials? Fiducials are frequently overlooked during PCB design. These small reference markers allow pick-and-place machines to identify the exact position and orientation of the PCB so that placement alignment can be corrected automatically. Without fiducials, your high-speed, high-precision SMT line might as well be playing darts blindfolded.

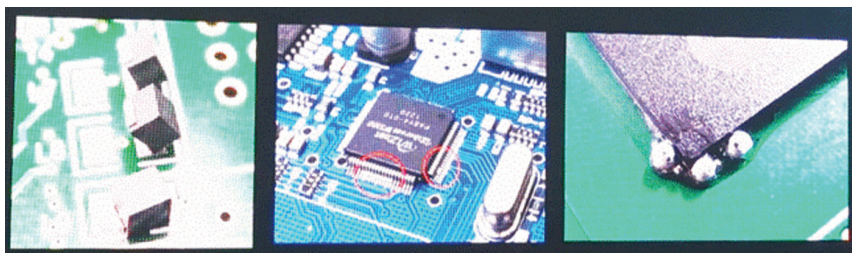


Fig. 2: Problems due to aperture defects

Fiducials and apertures may appear to be minor layout details, yet they play a critical role in achieving smooth, defect-free PCB assembly. Skipping them—or implementing them incorrectly—can introduce costly manufacturing problems.

Panel fiducial. This is for the panel itself and is primarily used during stencil printing or paste printing. Think of it as the machine's 'home base' marker.

Global PCB fiducial. Even if you have got multiple boards on a panel, each PCB should have its own set of fiducials. These act as individual 'anchors' for alignment.

Local fiducial. When working with small or high-precision components—say, a BGA processor like the one in a Raspberry Pi—you need local fiducials near that part. These are especially critical for packages like QFN or BGA, where even the slightest misalignment can cause shorts or other failures.

For instance, the two small golden dots commonly seen beside a BGA component serve as local fiducials, ensuring the integrated circuit is placed accurately during assembly.

If fiducials provide the positional reference for assembly, apertures control how much solder paste is deposited on each pad. In PCB design, aperture optimisation is often overlooked.

Before assembly begins, the PCB undergoes stencil printing, where solder paste is applied through laser-cut openings in a stencil known as apertures. Component datasheets typically specify recommended

aperture designs. Ignoring these specifications can lead to excessive solder paste deposition. The result?

Tombstoning. Components like inductors or tiny 0201/0402 packages lifting off one pad and standing upright

Bridging. Solder paste forming unintended connections between pins, even on large TQFP packages

BGA balling. Excess paste causing hidden solder balls under BGAs, only detectable via X-ray

Thousands of boards are plagued by these issues, and the cost of identifying and fixing them post-production can be eye-watering. In the high-speed, high-stakes world of electronics manufacturing, prevention is always cheaper than cure. A robust, regularly reviewed BoM is more than a purchasing checklist—it is your insurance against sourcing nightmares, compliance lapses, and assembly line disasters. Marrying cost efficiency with technical foresight, compliance discipline, and manufacturing practicality transforms your BoM from a static document into a living, evolving strategy.

Factor in the often-overlooked heroes—from fiducials to moisture control—and you will not only cut costs but also safeguard quality, reliability, and delivery timelines. Because in this industry, it's not just the big decisions that define success—it is the thousands of small ones you get right before the first board rolls off the line. **EFY**

This article is based on a tech talk session at IEW 2025 in Bengaluru by Srinivas Parikshith, CEO, Auckam Technologies. It was transcribed and curated by Akanksha Sondhi Gaur, Sr Technical Journalist at EFY.